

Human Development Index (HDI) Predictor

Functional Features

Project: Human Development Index (HDI) Predictor
Phase: Project Development Phase

1. Core Features

Feature	Description
Country dropdown	Pre-populated from the trained encoder's known country classes
Indicator input fields	Life Expectancy, Mean Years of Schooling, GNI per Capita, Internet Users with sensible min/max ranges and placeholders
HDI score prediction	Server-side prediction via the trained Linear Regression model, clamped to a valid [0, 1] range
Development tier classification	Automatic mapping to Low / Medium / High / Very High Human Development
Result display	Styled result page showing the score, category badge, and country name
Predict again	One-click return to the prediction form to test another country

2. Supporting Features

- Home page explaining what the HDI is and how the tool works
- Graceful error handling for invalid or incomplete form submissions
- Consistent dark-themed styling across all three pages